



Selenium End-to-End Testing — Homework Report

WEBSITE UNDER TEST

practicesoftwaretesting.com

Name Biruk Mulatu

ID ATE/9686/14

Submitted for Abel Tadesse

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1. Site chosen and why

practicesoftwaretesting.com — a demo tool shop (Toolshop) built and maintained specifically for test-automation practice. It provides a real login flow, a product catalogue with search, an add-to-cart-and-checkout flow, and both client-side and server-side form validation, satisfying the assignment's requirement for a form/login/search feature with a visible result and a way to trigger an error. Every element also carries a stable data-test attribute, since the site is purpose-built for this kind of testing.

2. Test case table

ID	Req.	Action	Input	Expected result	Actual result	Result
TC01	T1	Open home page	practicesoftwaretesting.com	Title = "Practice Software Testing - Toolshop - v5.0"; catalogue renders ≥ 1 product	Title matched; catalogue rendered	Pass
TC02	T3	Log in with valid credentials	customer@practicesoftwaretesting.com / welcome01	Redirected to /account; title "My account"; nav shows user's name	Landed on /account; title matched; nav showed "Jane Doe"	Pass
TC03	T3	Search a product, open it, add to cart, view cart	Search term "Hammer"	Search returns "Hammer"; product page shows it; cart badge = 1; cart total $> \$0$	Product found and added; badge = 1; cart showed "Hammer", total = \$12.58	Pass
TC04	T4	Log in with wrong password	customer@practicesoftwaretesting.com / definitely_wrong	Error "Invalid email or password"; no redirect	Error shown exactly; stayed on /auth/login	Pass
TC05	T4	Submit login form with both fields empty	"" / ""	"Email is required" and "Password is required" shown; form blocked	Both messages shown; no redirect	Pass
TC06	T4	Search a nonsense term	zzzznoresultsxyz123	"No products found" message; 0 product tiles	Message shown; 0 tiles rendered	Pass
TC07	T6 (P1)	Parameterized login — valid email + correct password	customer@practicesoftwaretesting.com / welcome01	Success, lands on /account	Landed on /account	Pass
TC08	T6 (P2)	Parameterized login — valid email + wrong password	customer@practicesoftwaretesting.com / wrong_password	Error "Invalid email or password"	Matched	Pass
TC09	T6 (P3)	Parameterized login — malformed email	not-an-email / welcome01	Error "Email format is invalid"	Matched	Pass
TC10	T6 (P4)	Parameterized login — both fields empty	"" / ""	Error "Email is required"	Matched	Pass

Cross-cutting (not separate cases): T2 — By.id and By.cssSelector used throughout, no positional XPath. T5 — explicit WebDriverWait/ExpectedConditions on every state transition (login redirect, search completion, an .or(...) across three possible validation alerts, and a custom lambda condition waiting for the cart badge to increment); no Thread.sleep anywhere. T7 — 5 Page Objects (LoginPage, AccountPage, ProductsPage, ProductDetailPage, CartPage); tests call only their methods. T8 — fresh ChromeDriver per test in @BeforeEach, driver.quit() in @AfterEach, headless by default.

3. Design technique for T6

Equivalence Partitioning on the login form's two inputs (email, password):

Partition	Input class	Expected outcome
P1	Well-formed, registered email + correct password	Success → /account
P2	Well-formed, registered email + wrong password	Server error: "Invalid email or password"
P3	Syntactically invalid email (no @)	Client error: "Email format is invalid"
P4	Both required fields empty	Client errors: "Email/Password is required"

4. Defects / odd behaviour found

- Cart page's product-title element has a trailing-whitespace text node before an Angular comment marker; ChromeDriver's `getText()` doesn't trim it consistently there (unlike the same element on the catalogue page) — worked around by trimming in the Page Object.
- The account nav menu (logged-in user's name) populates slightly after the rest of the "My account" page loads — a separate async call, so waiting only on the page title is a race condition.
- Navigating to `/auth/login` while already authenticated does not redirect away from the login form.
- Under repeated rapid login attempts (~10+ within 15 minutes, during test development) the backend began returning a generic "Login failed" message instead of the specific "Invalid email or password", and a valid login stopped redirecting — consistent with informal rate-limiting. Resolved by spacing out test runs, per the assignment's "run a handful of times" rule.

5. Evidence of a run — mvn test console output

```
[INFO] --- maven-surefire-plugin:3.0.0:test (default-test) @ selenium-homework ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running et.edu.aau.selenium.tests.LoginPositiveTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 9.918 s -- in et.edu.aau.selenium.tests.LoginPositiveTest
[INFO] Running et.edu.aau.selenium.tests.LoginDataDrivenTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 8.207 s -- in et.edu.aau.selenium.tests.LoginDataDrivenTest
[INFO] Running et.edu.aau.selenium.tests.LoginNegativeTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 3.876 s -- in et.edu.aau.selenium.tests.LoginNegativeTest
[INFO] Running et.edu.aau.selenium.tests.SearchNegativeTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 2.115 s -- in et.edu.aau.selenium.tests.SearchNegativeTest
[INFO] Running et.edu.aau.selenium.tests.SmokeTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.505 s -- in et.edu.aau.selenium.tests.SmokeTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 28.136 s
[INFO] Finished at: 2026-08-30T14:18:32+03:00
[INFO] -----
```